

PAVAN KUMAR

Bengaluru | pavanram2000@gmail.com | 9035526622 | LinkedIn | Github

Profile Summary

Data Engineer with **3+ years** of experience designing and optimizing large-scale **PySpark and Spark SQL** ETL pipelines on **AWS** and **Azure Databricks**, processing **100+ TB/day** to power customer targeting campaigns across 100+ countries generating **200M+ annual revenue**. Hands-on experience in data quality, anomaly detection, observability, and **LLM-based** metadata generation.

Tools & Technologies

Programming: Python, SQL, Shell Scripting

Big Data & Processing: Apache Spark, PySpark, Spark SQL, Delta Lake, Databricks

Cloud & Data Platforms: AWS (S3, EMR, Athena)

Orchestration: Apache Airflow

Data Engineering: ETL/ELT Pipelines, Medallion Architecture, Data Modeling, Batch Processing, Data Quality, Lakehouse Architecture

AI & GenAI: Fundamentals of Machine learning, LLMs, RAG, Vector Databases, NLP

DevOps & Tools: Git, CI/CD Pipelines, JIRA, Agile/Scrum

Storage & Formats: Parquet, Delta

Work Experience

Deloitte USI – AWS Data Engineer (August 2022 – Present)

- Built and optimized large-scale **PySpark and Spark SQL** ETL pipelines on AWS EMR orchestrated via Apache Airflow, processing **100+ TB** of data daily to power customer targeting campaigns across 100+ countries and 500K+ weekly users, contributing to business initiatives generating **200M+ annual revenue**.
- Collaborated with business and product teams to translate targeting requirements into pipeline enhancements, extending data models with new attributes to enable machine-level targeting across all segments, increasing targeting rate by **10%** per quarter.
- As part of a cross-platform observability initiative, implemented anomaly detection using **Azure Databricks Lakehouse** Monitoring across **8 crucial tables**, flagging metric deviations by comparing current values against historical averages using configurable thresholds — enabling early notification to business stakeholders before issues impact downstream analytics.
- Supported and extended an **LLM-based metadata generation** application on **Azure Databricks** that auto-generates table column descriptions from wikis, Git repositories, codebases, and KT transcripts — onboarding **10+ tables**, updating DDLs, and troubleshooting pipeline failures.
- Leveraged **AWS Athena** to address ad-hoc business questions and enable analysis of large-scale datasets, providing rapid insights to stakeholders that accelerated critical business decisions and reduced reporting turnaround times.
- Configured automated Airflow status reports every 3 hours consolidating all DAG execution statuses segregated by process, reducing manual monitoring by **2 hours** daily and enabling proactive data quality issue resolution.

Certifications and Awards

- AWS Certified Developer Associate - Credential Verification
- AWS Certified Machine Learning Specialty - Credential Verification
- Databricks Certified Data Engineer Associate - Credential Verification
- Recognized with 2 Outstanding and 3 Applause Awards for exceptional performance at Deloitte USI.

Education

Christ University — B.Sc. Computer Science
St. Francis P.U. College — PCME

CGPA: 8.42/10